

Homework #8

The Gaussian Channel

Introduction to Information Theory 2026 · EC5203 / EC4310 / MM4450 · Prof. Heung-No Lee · GIST

How to submit. Put your solution in the `submission/` folder of your assignment repository: a single PDF (`submission/hw8.pdf` — scanned handwriting is fine if legible), or Markdown with L^AT_EX (`submission/hw8.md`). Commit and push before the deadline (23:59 KST); the last commit before the deadline is graded. If you worked as a group (max 3), list all members at the top — each member still submits in their own repository.

Questions. Light ones in the AI TA chat on the course homepage; rigorous derivations and proofs on the course GitHub Q&A (mention @claude). Before the deadline you get hints, never solutions.

Problem HW8.1 (C&T 8.1(c)) *Differential entropy.* Evaluate the differential entropy $h(X) = -\int f \ln f$ for the following:

- (c) The sum of X_1 and X_2 , where X_1 and X_2 are independent normal random variables with means μ_i and variances σ_i^2 , $i = 1, 2$.

Problem HW8.2 (C&T 8.5) *Scaling.* Let $h(\mathbf{X}) = -\int f(\mathbf{x}) \log f(\mathbf{x}) d\mathbf{x}$. Show

$$h(A\mathbf{X}) = \log |\det(A)| + h(\mathbf{X}).$$

Hint: Since $\mathbf{y} = A\mathbf{x}$, the Jacobian matrix is simply A . Therefore,

$$g(\mathbf{y}) = \frac{f(A^{-1}\mathbf{y})}{|\det(A)|}.$$

Now obtain the differential entropy of $\mathbf{y}$.

Problem HW8.3 (C&T 9.2) *Two-look Gaussian channel.*



Consider the ordinary Gaussian channel with two correlated looks at X , that is, $Y = (Y_1, Y_2)$, where

$$Y_1 = X + Z_1 \tag{9.171}$$

$$Y_2 = X + Z_2 \tag{9.172}$$

with a power constraint P on X , and $(Z_1, Z_2) \sim \mathcal{N}_2(0, K)$, where

$$K = \begin{bmatrix} N & N\rho \\ N\rho & N \end{bmatrix}. \tag{9.173}$$

Find the capacity C for

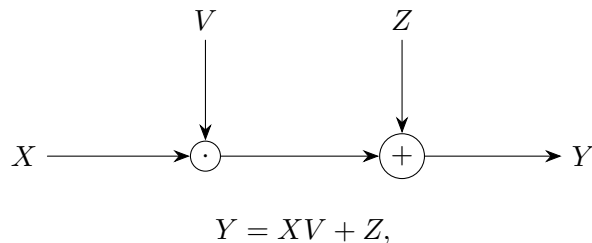
(a) $\rho = 1$

(b) $\rho = 0$

(c) $\rho = -1$

Problem HW8.4 (C&T 9.3) *Output power constraint.* Consider an additive white Gaussian noise channel with an expected *output* power constraint P . Thus, $Y = X + Z$, $Z \sim \mathcal{N}(0, \sigma^2)$, Z is independent of X , and $\mathbb{E}Y^2 \leq P$. Find the channel capacity.

Problem HW8.5 (C&T 9.5) *Fading channel.* Consider an additive noise fading channel



where Z is additive noise, V is a random variable representing fading, and Z and V are independent of each other and of X . Argue that knowledge of the fading factor V improves capacity by showing that

$$I(X; Y|V) \geq I(X; Y).$$

Problem HW8.6 (C&T 8.2/8.9/9.12/9.20) *Additional assigned problems.* From the slides' assignment, solve also 8.2 (concavity of determinants), 8.9 (Gaussian mutual information along a Markov chain), 9.12 (time-varying channel: show via Fano that the capacity is zero), and 9.20 (robust decoding with non-Gaussian noise).

Problem HW8.7 *Units of differential entropy.* In Example 8.3 of the Course Book (Gaussian density) the differential entropy of $X \sim \mathcal{N}(0, \sigma^2)$ was computed in nats as $h(X) = \frac{1}{2} \ln(2\pi\sigma^2 e)$. Using $\log_e A = \log_2 A / \log_2 e$, express $h(X)$ in bits, and evaluate it for $\sigma^2 = 1000$.

Problem HW8.8 *Equality in the capacity derivation.* In the proof of Theorem 8.15 of the Course Book (capacity of the Gaussian channel) we bounded $I(X; Y) \leq \frac{1}{2} \log 2\pi e(P + N) - \frac{1}{2} \log 2\pi eN$. When do we have the equality? State the condition on the distribution of Y , and explain why the input $X \sim \mathcal{N}(0, P)$ meets it.